

1/ part 1a

$$\pi(x) = c e^{-\lambda|x|}, \quad x \in \mathbb{R}$$

$$\int_{\mathbb{R}} \pi(x) dx = 1 \Rightarrow c \int_{\mathbb{R}} e^{-\lambda|x|} dx = 1$$

$$\Rightarrow 2c \int_0^{\infty} e^{-\lambda x} dx = 1 \quad (\text{by symmetry of } |x| \text{ around } 0)$$

$$\Rightarrow 2c \left. \frac{e^{-\lambda x}}{-\lambda} \right|_0^{\infty} = 1 \Rightarrow 2c \cdot \frac{0 - (+1)}{-\lambda} = 1$$

$$\Rightarrow 2c \cdot \frac{1}{\lambda} = 1 \Rightarrow c = \lambda/2. \quad \square$$

part 1b

$$\pi(x) = c e^{-\lambda x^2}, \quad x \in \mathbb{R}$$

$$\int_{\mathbb{R}} \pi(x) dx = 1 \Rightarrow c \int_{\mathbb{R}} e^{-\lambda x^2} dx = 1$$

$$\Rightarrow c \cdot \sqrt{\pi/\lambda} \int \frac{1}{\sqrt{2\pi \cdot \frac{1}{2\lambda}}} \cdot e^{-\frac{1}{2} \cdot \frac{1}{\frac{1}{2\lambda}} x^2} dx = 1$$

$$\Rightarrow c \sqrt{\pi/\lambda} \times 1 = 1 \Rightarrow c = \sqrt{\lambda/\pi}. \quad \square$$

In the second part,

$$y_i | \beta \stackrel{\text{indep}}{\sim} \text{Normal}(\beta x_i, 1)$$

$$\begin{aligned} \text{Likelihood} &= L(\beta | y_1, \dots, y_n) \propto \prod_{i=1}^n e^{-\frac{1}{2}(y_i - \beta x_i)^2} \\ &\propto e^{-\frac{1}{2} \sum_{i=1}^n (y_i - \beta x_i)^2} \end{aligned}$$

Part 2a

$$\text{Prior: } \pi(\beta) = \frac{\lambda}{2} e^{-\lambda|\beta|}$$

Thus, posterior $\propto$ Likelihood $\times$ Prior

$$\propto e^{-\frac{1}{2} \sum_{i=1}^n (y_i - \beta x_i)^2} \times e^{-\lambda|\beta|}$$
$$e^{-\frac{1}{2} \left[\sum_{i=1}^n (y_i - \beta x_i)^2 + 2\lambda|\beta| \right]}$$

Thus, the MAP estimator is obtained by maximizing the posterior w.r.t. β , i.e.,

$$\hat{\beta}_{\text{MAP}} = \underset{\beta}{\operatorname{argmax}} e^{-\frac{1}{2} \left[\sum_{i=1}^n (y_i - \beta x_i)^2 + 2\lambda|\beta| \right]}$$
$$= \underset{\beta}{\operatorname{argmin}} \left(\sum_{i=1}^n (y_i - \beta x_i)^2 + 2\lambda|\beta| \right) \quad (1)$$

In this context, the LASSO estimate would satisfy

$$\hat{\beta}_{\text{Lasso}} = \underset{\beta}{\operatorname{argmin}} \left(\sum_{i=1}^n (y_i - \beta x_i)^2 + \lambda^*|\beta| \right) \quad (2)$$

for the tuning parameter λ^* .

By Comparing (1) and (2), we see that

$\hat{\beta}_{\text{MAP}}$ coincides with $\hat{\beta}_{\text{Lasso}}$ when $2\lambda = \lambda^*$.

Part 2b

$$\text{Prior: } \pi(\beta) = \sqrt{\frac{\pi}{\lambda}} e^{-\lambda \beta^2}$$

Thus, posterior $\propto$ Likelihood $\times$ Prior

$$\propto e^{-\frac{1}{2} \sum_{i=1}^n (y_i - \beta x_i)^2} \times e^{-\lambda \beta^2}$$
$$e^{-\frac{1}{2} \left[\sum_{i=1}^n (y_i - \beta x_i)^2 + 2\lambda \beta^2 \right]}$$

Thus, the MAP estimator is obtained by maximizing the posterior w.r.t. β , i.e.,

$$\hat{\beta}_{\text{MAP}} = \underset{\beta}{\operatorname{argmax}} e^{-\frac{1}{2} \left[\sum_{i=1}^n (y_i - \beta x_i)^2 + 2\lambda \beta^2 \right]}$$
$$= \underset{\beta}{\operatorname{argmin}} \left(\sum_{i=1}^n (y_i - \beta x_i)^2 + 2\lambda \beta^2 \right) \quad (1)$$

In this context, the ridge estimate would satisfy

$$\hat{\beta}_{\text{Lasso}} = \underset{\beta}{\operatorname{argmin}} \left(\sum_{i=1}^n (y_i - \beta x_i)^2 + \lambda^* \beta^2 \right) \quad (2)$$

for the tuning parameter λ^* .

By Comparing (1) and (2), we see that

$\hat{\beta}_{\text{MAP}}$ coincides with $\hat{\beta}_{\text{ridge}}$ when $2\lambda = \lambda^*$.

$$2 / X_1 \sim \text{Normal}(0, \sigma^2)$$

$$X_t | X_{t-1} \sim \text{Normal}(\rho X_{t-1}, \sigma^2(1-\rho^2))$$

$$\text{prior: } \sigma^2 \sim \text{IG}(0.1, 0.1)$$

ρ is known.

$$\text{Likelihood} = L(\sigma^2) = \pi(X_1, \dots, X_T | \sigma^2)$$

$$\text{posterior } \pi(X_1, \dots, X_T) \propto \text{Likelihood} \times \text{prior}$$

$$\propto \pi(X_1 | \sigma^2) \pi(X_2 | X_1, \sigma^2) \dots \pi(X_T | X_{T-1}, \sigma^2) \times \pi(\sigma^2)$$

$$\propto \frac{1}{\sigma} e^{-\frac{1}{2} \frac{X_1^2}{\sigma^2}} \times \frac{1}{\sigma} e^{-\frac{1}{2} \frac{(X_2 - \rho X_1)^2}{\sigma^2(1-\rho^2)}} \times \dots \times \frac{1}{\sigma} e^{-\frac{1}{2} \frac{(X_T - \rho X_{T-1})^2}{\sigma^2(1-\rho^2)}} \times (\sigma^2)^{-0.1} e^{-\frac{0.1}{\sigma^2}}$$

$$\propto (\sigma^2)^{-\left(\frac{T}{2} + 0.1\right)} e^{-\frac{1}{\sigma^2} \left[\frac{1}{2} X_1^2 + \frac{1}{2} \frac{(X_2 - \rho X_1)^2}{1-\rho^2} + \dots + \frac{1}{2} \frac{(X_T - \rho X_{T-1})^2}{1-\rho^2} + 0.1 \right]}$$

$$\sigma^2 | X_1, \dots, X_T \sim \text{IG}\left(\frac{T}{2} + 0.1, \frac{1}{2} X_1^2 + \frac{1}{2(1-\rho^2)} \sum_{t=2}^T (X_t - \rho X_{t-1})^2 + 0.1\right)$$

$$E(\sigma^2 | X_1, \dots, X_T) = \frac{\frac{1}{2} X_1^2 + \frac{1}{2(1-\rho^2)} \sum_{t=2}^T (X_t - \rho X_{t-1})^2 + 0.1}{\frac{T}{2} + 0.1 - 1}$$

$$V(\sigma^2 | X_1, \dots, X_T) = \frac{\left\{ \frac{1}{2} X_1^2 + \frac{1}{2(1-\rho^2)} \sum_{t=2}^T (X_t - \rho X_{t-1})^2 + 0.1 \right\}^2}{\left(\frac{T}{2} + 0.1 - 1 \right)^2 \left(\frac{T}{2} + 0.1 - 2 \right)}$$

Once we observe X_{T+1} , we know that

$$E(\sigma^2 | X_1, \dots, X_T, X_{T+1})$$

$$= \frac{\frac{1}{2} X_1^2 + \frac{1}{2(1-\rho^2)} \sum_{t=2}^{T+1} (X_t - \rho X_{t-1})^2 + 0.1}{\frac{T+1}{2} + 0.1 - 1}$$

$$= \frac{\left(\frac{T}{2} + 0.1 - 1\right) \times E(\sigma^2 | X_1, \dots, X_T) + \frac{1}{2(1-\beta^2)} (X_{T+1} - \beta X_T)^2}{\frac{T+1}{2} + 0.1 - 1}$$

$$V(\sigma^2 | X_1, \dots, X_T, X_{T+1})$$

$$= \frac{\left\{ \frac{1}{2} X_1^2 + \frac{1}{2(1-\beta^2)} \sum_{t=2}^{T+1} (X_t - \beta X_{t-1})^2 + 0.1 \right\}^2}{\left(\frac{T+1}{2} + 0.1 - 1\right)^2 \left(\frac{T+1}{2} + 0.1 - 2\right)}$$

$$= \frac{\left\{ \left(\frac{T}{2} + 0.1 - 1\right) \left(\frac{T}{2} + 0.1 - 2\right)^{\frac{1}{2}} \sqrt{V(\sigma^2 | X_1, \dots, X_T)} + \frac{1}{2(1-\beta^2)} (X_{T+1} - \beta X_T)^2 \right\}^2}{\left(\frac{T+1}{2} + 0.1 - 1\right)^2 \left(\frac{T+1}{2} + 0.1 - 2\right)}$$

Thus, once we know $E(\sigma^2 | X_1, \dots, X_T)$ and $V(\sigma^2 | X_1, \dots, X_T)$, we can calculate $E(\sigma^2 | X_1, \dots, X_T, X_{T+1})$ and $V(\sigma^2 | X_1, \dots, X_T, X_{T+1})$ after only a few calculations.

$$3/ \quad \tilde{X}_1 \sim N_p(\tilde{\mu}, \tilde{\Sigma})$$

$$\tilde{X}_t | \tilde{X}_{t-1} \sim N_p(\beta \tilde{X}_{t-1}, \tilde{\Sigma}), \quad t = 2, \dots, T$$

In this question, β is known and $\tilde{\Sigma}$ is unknown.

$$\tilde{\Sigma} \sim \text{Inverse-Wishart}(\mathbf{I}_p, \nu)$$

$$\pi(\Sigma | \underline{x}_1, \dots, \underline{x}_T) \propto \pi(\underline{x}_1, \dots, \underline{x}_T | \Sigma) \pi(\Sigma)$$

$$\propto \pi(\underline{x}_1 | \Sigma) \pi(\underline{x}_2 | \underline{x}_1, \Sigma) \times \dots \times \pi(\underline{x}_T | \underline{x}_{T-1}, \Sigma) \times \pi(\Sigma)$$

$$\propto \frac{1}{|2\pi\Sigma|^{1/2}} e^{-\frac{1}{2} \underline{x}_1' \Sigma^{-1} \underline{x}_1} \times \frac{1}{|2\pi\Sigma|^{1/2}} e^{-\frac{1}{2} (\underline{x}_2 - \beta \underline{x}_1)' \Sigma^{-1} (\underline{x}_2 - \beta \underline{x}_1)} \times \dots$$

$$\times \frac{1}{|2\pi\Sigma|^{1/2}} e^{-\frac{1}{2} (\underline{x}_T - \beta \underline{x}_{T-1})' \Sigma^{-1} (\underline{x}_T - \beta \underline{x}_{T-1})}$$

$$\times \frac{|\mathbf{I}|^{+v/2}}{2^{vp/2} \Gamma_p(v/2)} \cdot |\Sigma|^{-(v+p+1)/2} e^{-\frac{1}{2} \text{Tr}(\mathbf{I} \Sigma^{-1})}$$

$$\propto |\Sigma|^{-\frac{T}{2}} e^{-\frac{1}{2} \left[\underline{x}_1' \Sigma^{-1} \underline{x}_1 + \sum_{t=2}^T (\underline{x}_t - \beta \underline{x}_{t-1})' \Sigma^{-1} (\underline{x}_t - \beta \underline{x}_{t-1}) \right]}$$

$$\times |\Sigma|^{-(v+p+1)/2} \cdot e^{-\frac{1}{2} \text{Tr}(\Sigma^{-1})}$$

$$\propto |\Sigma|^{-(v+T+p+1)/2} \cdot e^{-\frac{1}{2} \text{Tr} \left(\left[\underline{x}_1 \underline{x}_1' + \sum_{t=2}^T (\underline{x}_t - \beta \underline{x}_{t-1}) (\underline{x}_t - \beta \underline{x}_{t-1})' + \mathbf{I} \right] \Sigma^{-1} \right)}$$

$$\Rightarrow \Sigma | \underline{x}_1, \dots, \underline{x}_T \sim \text{IW} \left(\left[\underline{x}_1 \underline{x}_1' + \sum_{t=2}^T (\underline{x}_t - \beta \underline{x}_{t-1}) (\underline{x}_t - \beta \underline{x}_{t-1})' + \mathbf{I} \right], T+v \right)$$

$$\text{So, } E\left(\sum \mid \underline{X}_1, \dots, \underline{X}_T\right) = \frac{\underline{X}_1 \underline{X}_1' + \sum_{t=2}^T (\underline{X}_t - \rho \underline{X}_{t-1})(\underline{X}_t - \rho \underline{X}_{t-1})' + \mathbf{I}}{T + \nu - p - 1}$$

if $T + \nu > p + 1$,

Otherwise $E\left(\sum \mid \underline{X}_1, \dots, \underline{X}_T\right)$ does not exist.

$$4/. \quad X_1, \dots, X_n \stackrel{iid}{\sim} \text{Exponential}(\lambda)$$

$$L(\lambda) = \pi(X_1, \dots, X_n; \lambda)$$

$$= \prod_{i=1}^n \lambda e^{-\lambda x_i} = \lambda^n e^{-\lambda \sum_{i=1}^n x_i}$$

$$\ell(\lambda) = \log L(\lambda) = n \log(\lambda) - \lambda \sum_{i=1}^n x_i$$

$$\frac{d\ell(\lambda)}{d\lambda} = \frac{n}{\lambda} - \sum_{i=1}^n x_i$$

$$\frac{d^2\ell(\lambda)}{d\lambda^2} = -\frac{n}{\lambda^2}$$

$$I(\lambda) = E\left(-\frac{d^2\ell(\lambda)}{d\lambda^2}\right) = \frac{n}{\lambda^2}$$

Thus, Jeffreys' prior $\pi(\lambda) \propto \sqrt{I(\lambda)} \propto \lambda^{-1}$

Now, $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Exponential}(\lambda)$

$$\begin{aligned}\pi(\lambda | x_1, \dots, x_n) &\propto \pi(x_1, x_2, \dots, x_n | \lambda) \pi(\lambda) \\ &\propto \left\{ \prod_{i=1}^n \lambda e^{-\lambda x_i} \right\} \times \lambda^{-1} \\ &\propto \lambda^{n-1} e^{-\lambda \sum_{i=1}^n x_i}\end{aligned}$$

Thus, $\lambda | x_1, \dots, x_n \sim \text{Gamma}(n, \sum_{i=1}^n x_i)$

MAP estimator of λ is

$$\begin{aligned}\arg\max_{\lambda} \pi(\lambda | x_1, \dots, x_n) &= \arg\max_{\lambda} \ln \pi(\lambda | x_1, \dots, x_n) \\ &= \arg\max_{\lambda} \left[(n-1) \log(\lambda) - \lambda \sum_{i=1}^n x_i \right]\end{aligned}$$

Thus, say $g(\lambda) = (n-1) \log(\lambda) - \lambda \sum_{i=1}^n x_i$

$$g'(\lambda) = (n-1) \frac{1}{\lambda} - \sum_{i=1}^n x_i$$

$$g''(\lambda) = -(n-1) \frac{1}{\lambda^2} < 0.$$

Thus, $g(\lambda)$ is maximum, when $g'(\lambda) = 0$

$$\Rightarrow \lambda = \frac{n-1}{\sum_{i=1}^n x_i}$$

$$\text{Thus } \hat{\lambda}_{\text{MAP}} = \frac{n-1}{\sum_{i=1}^n x_i} = \frac{1 - 1/n}{\bar{x}}.$$

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Exponential}(\lambda)$$

$$\Rightarrow \sum_{i=1}^n X_i \sim \text{Gamma}(n, \lambda)$$

$$\Rightarrow \bar{X} \sim \text{Gamma}(n, n\lambda)$$

$$\Rightarrow \frac{1}{\bar{X}} \sim \text{Inverse-gamma}(n, n\lambda).$$

$$\Rightarrow E\left(\frac{1}{\bar{X}}\right) = \frac{n\lambda}{n-1} = \frac{\lambda}{1 - 1/n}$$

$$\Rightarrow E\left(\frac{1 - 1/n}{\bar{X}}\right) = \lambda$$

$$\Rightarrow E(\hat{\lambda}_{\text{MAP}}) = \lambda.$$

$$V\left(\frac{1}{\bar{X}}\right) = \frac{(n\lambda)^2}{(n-1)^2(n-2)} = \frac{\lambda^2}{\left(1 - \frac{1}{n}\right)^2(n-2)}$$

$$\Rightarrow V\left(\frac{1 - 1/n}{\bar{X}}\right) = \frac{\lambda^2}{n-2}.$$

$$X_{n+1} | \lambda \sim \text{Exp}(\lambda)$$

$$\lambda | x_1, \dots, x_n \sim \text{Ga}\left(n, \sum_{i=1}^n x_i\right)$$

$$\begin{aligned} \pi(x_{n+1} | x_1, \dots, x_n) &= \int \pi(x_{n+1} | \lambda) \pi(\lambda | x_1, \dots, x_n) d\lambda \\ &= \int \lambda e^{-\lambda x_{n+1}} \frac{\left(\sum_{i=1}^n x_i\right)^n}{\Gamma(n)} \cdot \lambda^{n-1} e^{-\lambda \sum_{i=1}^n x_i} d\lambda \\ &= \frac{\left(\sum_{i=1}^n x_i\right)^n}{\Gamma(n)} \int \lambda^{n+1-1} e^{-\lambda \sum_{i=1}^{n+1} x_i} d\lambda \\ &= \frac{\left(\sum_{i=1}^n x_i\right)^n}{\Gamma(n)} \times \frac{\Gamma(n+1)}{\left(\sum_{i=1}^{n+1} x_i\right)^{n+1}} \times \\ &\quad \int \frac{\left(\sum_{i=1}^{n+1} x_i\right)^{n+1}}{\Gamma(n+1)} \cdot \lambda^{n+1-1} e^{-\lambda \sum_{i=1}^{n+1} x_i} d\lambda \\ &= \frac{\Gamma(n+1)}{\Gamma(n)} \cdot \frac{\left(\sum_{i=1}^n x_i\right)^n}{\left(\sum_{i=1}^{n+1} x_i\right)^{n+1}} \\ &= n \frac{\left(\sum_{i=1}^n x_i\right)^n}{\left(\sum_{i=1}^{n+1} x_i\right)^{n+1}} \end{aligned}$$

5/. Let $\underline{y}_t$ be the vector denoting the wins, loses and draws during the t th season. Let $\underline{y}_t$ be the data

$$\underline{y}_1 = (4, 4, 2)', \underline{y}_2 = (5, 3, 2)'$$

$$\underline{\theta} = (\theta_1, \theta_2, \theta_3)' \sim \text{Dir}(1, 1, 1)$$

$$\underline{y}_1 | \underline{\theta} \sim \text{Multinomial}(10; \theta_1, \theta_2, \theta_3)$$

$$\underline{y}_2 | \underline{\theta} \sim \text{Multinomial}(10; \theta_1, \theta_2, \theta_3)$$

our target is to calculate

$$Pr(\underline{y}_3 = \underline{y}_3 | \underline{y}_1 = \underline{y}_1, \underline{y}_2 = \underline{y}_2)$$

θ_1 = win probability

θ_2 = lose probability

θ_3 = draw probability

$$\text{where } \underline{y}_3 = (9, 1, 0)'$$

$$Pr(\underline{y}_3 = \underline{y}_3 | \underline{y}_1 = \underline{y}_1, \underline{y}_2 = \underline{y}_2)$$

$$= \int_S P(\underline{y}_3 = \underline{y}_3 | \underline{\theta}) \pi(\underline{\theta} | \underline{y}_1 = \underline{y}_1, \underline{y}_2 = \underline{y}_2) d\underline{\theta}$$

where S denotes the 2-D Simplex

$$S = \left\{ (\theta_1, \theta_2, \theta_3) : 0 < \theta_1, \theta_2, \theta_3 < 1, \sum_{i=1}^3 \theta_i = 1 \right\}$$

$$\begin{aligned}
\text{Now, } \pi(\underline{\theta} | \underline{y}_1 = \underline{y}_1, \underline{y}_2 = \underline{y}_2) &\propto \pi(\underline{y}_1 = \underline{y}_1, \underline{y}_2 = \underline{y}_2 | \underline{\theta}) \pi(\underline{\theta}) \\
&\propto \pi(\underline{y}_1 = \underline{y}_1 | \underline{\theta}) \pi(\underline{y}_2 = \underline{y}_2 | \underline{\theta}) \pi(\underline{\theta}) \\
&\propto \theta_1^5 \theta_2^3 \theta_3^2 \times \theta_1^4 \theta_2^4 \theta_3^2 \times \mathbb{1}(\underline{\theta} \in S) \\
&\propto \theta_1^9 \theta_2^7 \theta_3^4 \times \mathbb{1}(\underline{\theta} \in S) \\
&\propto \theta_1^{10-1} \theta_2^{8-1} \theta_3^{5-1} \mathbb{1}(\underline{\theta} \in S)
\end{aligned}$$

$$\Rightarrow \underline{\theta} | \underline{y}_1 = \underline{y}_1, \underline{y}_2 = \underline{y}_2 \sim \text{Dir}(10, 8, 5)$$

Thus,

$$\begin{aligned}
\pi(\underline{\theta} | \underline{y}_1 = \underline{y}_1, \underline{y}_2 = \underline{y}_2) &= \frac{\Gamma(10+8+5)}{\Gamma(10)\Gamma(8)\Gamma(5)} \cdot \theta_1^9 \theta_2^7 \theta_3^4 \times \mathbb{1}(\underline{\theta} \in S) \\
&= \frac{\Gamma(23)}{\Gamma(10)\Gamma(8)\Gamma(5)} \cdot \theta_1^9 \theta_2^7 \theta_3^4 \times \mathbb{1}(\underline{\theta} \in S)
\end{aligned}$$

Thus, $P(\underline{Y}_3 = \underline{y}_3 | \underline{Y}_1 = \underline{y}_1, \underline{Y}_2 = \underline{y}_2)$, Let $\underline{y}_3 = (a, b, c)'$

$$= \int_S \frac{10!}{a! b! c!} \theta_1^a \theta_2^b \theta_3^c \times \frac{\Gamma(23)}{\Gamma(10)\Gamma(8)\Gamma(5)} \cdot \theta_1^9 \theta_2^7 \theta_3^4 d\theta_1 d\theta_2 d\theta_3$$

$$= \frac{10!}{a! b! c!} \times \frac{\Gamma(23)}{\Gamma(10)\Gamma(8)\Gamma(5)} \int_S \theta_1^{a+9} \theta_2^{b+7} \theta_3^{c+4} d\theta_1 d\theta_2 d\theta_3$$

$$= \frac{10!}{a! b! c!} \times \frac{\Gamma(23)}{\Gamma(10)\Gamma(8)\Gamma(5)} \times \frac{\Gamma(a+10)\Gamma(b+8)\Gamma(c+5)}{\Gamma(a+b+c+23)}$$

Thus, for $\underline{y}_3 = (9, 1, 0)'$

$$Pr(\underline{Y}_3 = \underline{y}_3 | \underline{Y}_1 = \underline{y}_1, \underline{Y}_2 = \underline{y}_2)$$

$$= \frac{10!}{9! 1! 0!} \times \frac{\Gamma(23)}{\Gamma(10)\Gamma(8)\Gamma(5)} \times \frac{\Gamma(19)\Gamma(9)\Gamma(5)}{\Gamma(33)}$$

$$= 10 \frac{\Gamma(23) \Gamma(19) \Gamma(9) \cancel{\Gamma(5)}}{\Gamma(10) \Gamma(8) \cancel{\Gamma(5)} \Gamma(33)}$$

$$\text{for } \underline{y}_4 = (8, 1, 1)'$$

$$P_n(\underline{y}_4 = \underline{y}_4 | \underline{y}_1 = \underline{y}_1, \underline{y}_2 = \underline{y}_2)$$

$$= \int_S P(\underline{y}_4 = \underline{y}_4 | \underline{\theta}) \pi(\underline{\theta} | \underline{y}_1 = \underline{y}_1, \underline{y}_2 = \underline{y}_2) d\underline{\theta}$$

$$= \frac{10!}{8! 1! 1!} \cdot \frac{\Gamma(23)}{\Gamma(10)\Gamma(8)\Gamma(5)} \times \frac{\Gamma(18)\Gamma(9)\Gamma(6)}{\Gamma(33)}$$

$$= 90 \frac{\Gamma(23)\Gamma(18)\Gamma(9)\Gamma(6)}{\Gamma(10)\Gamma(8)\Gamma(5)\Gamma(33)}$$

$$\text{Here, } (\theta_1, \theta_2, \theta_3)' | \underline{y}_1, \underline{y}_2 \sim \text{Dirichlet}(10, 8, 5)$$

$$\Rightarrow \left(\underset{\theta^*}{\theta_1 + \theta_3}, \underset{1 - \theta^*}{\theta_2} \right) | \underline{y}_1, \underline{y}_2 \sim \text{Beta}(15, 8)$$

θ^* = probability of either win or loss.

$$\underline{y}_5 = (y_5^{(\text{win})}, y_5^{(\text{lose})}, y_5^{(\text{Draw})})$$

$$Z_5 = y_5^{(\text{win})} + y_5^{(\text{draw})}$$

$$Z_5 \sim \text{Bin}(10, \theta^*)$$

$$p_n(z_5 = 8 \mid \tilde{y}_1, \tilde{y}_2)$$

$$= \int \binom{10}{8} \theta^{*8} (1-\theta^*)^2 \times \frac{\Gamma(23)}{\Gamma(15)\Gamma(8)} \cdot \theta^{*15-1} (1-\theta^*)^{8-1} d\theta^*$$

$$= \binom{10}{8} \frac{\Gamma(23)}{\Gamma(18)\Gamma(5)} \int \theta^{*23-1} (1-\theta^*)^{10-1} d\theta^*$$

$$= \frac{10!}{8!2!} \cdot \frac{\Gamma(23)}{\Gamma(18)\Gamma(5)} \times \frac{\Gamma(23)\Gamma(10)}{\Gamma(33)}$$

$$= 45 \frac{\Gamma(23)\Gamma(23)\Gamma(10)}{\Gamma(18)\Gamma(5)\Gamma(33)} \cdot$$